

ALTER vs UPDATE

DBMS Revision Notes & Quick Reference

BYTE BATTLE / QUIZ PREP

1. Basic Difference ★★☆☆

ALTER (Structure)	UPDATE (Data)
DDL (Data Definition Language) command	DML (Data Manipulation Language) command
Works on structure / schema	Works on data / records
Changes the overall design of a table	Changes values stored inside a table
Can add, drop, or modify columns	Can change specific column values
<i>Example:</i> Add a new column to table	<i>Example:</i> Increase salary of an employee

🧠 Easy Memory Trick:

ALTER → Modifies Table Structure | **UPDATE** → Modifies Table Data

2. Base Example: Employee Table

STRUCTURE DEFINITION

- **Table Name:** Employee
- **Columns:** ID, Name, Salary
- **Data Types:** INT, VARCHAR, INT

STORED DATA CONTENT

ID	Name	Salary
1	A	10000
2	B	20000
3	C	30000

3. What can ALTER do?

Add a Column

```
ALTER TABLE Employee
ADD Email VARCHAR2(30);
```

Resulting Structure: ID | Name | Salary | Email

Delete a Column

```
ALTER TABLE Employee
DROP COLUMN Name;
```

Name column is permanently deleted from schema.

Change Datatype or Size

4. What does UPDATE do?

Modifies existing data values inside rows without changing table structure.

Basic Syntax

```
UPDATE table_name
SET column_name = value;
```

Example: Bulk Update

```
UPDATE Employee
SET Salary = Salary * 2;
```

Doubles salary for **every** employee in the table.

ID	Name	Old Salary	New Salary
----	------	------------	------------